WG01

Physical Geography: Mountains, Plateaus, Plains, Deserts

About 1-2 questions a paper. Two things repeat: highest peak by continent (RAS 2016, 2018, 2021) and desert matched to continent (RAS 2024). Everything else is a single one-liner.

World Geography carries only about 3 questions in the whole paper, and this chapter supplies most of them. Almost nothing here is analytical - it is peaks, deserts, winds and currents matched to places. Learn it as pairs, not as prose.

Physical Geography of the World

Interior & tectonics
- Moho
- Gutenberg
- Pangaea
- Ring of Fire

Mountains
- Everest
- Aconcagua
- Denali
- Kilimanjaro
- Elbrus
- Vinson

Plateaus & plains
- Tibet
- Colorado
- Katanga
- Anatolian
- Great Plains

Deserts
- Sahara
- Gobi
- Atacama
- Kalahari
- Great Victoria
- Patagonian

Landforms
- Barchan
- Yardang
- Cirque
- Fjord
- Drumlin

Air & ocean
- Pressure belts
- Local winds
- Currents
- Tides
- Coral reefs

Inside the earth, and why the continents move

Start with what the earth is made of, because the discontinuities and the wave behaviour are asked as straight one-liners. The earth has three shells - crust, mantle, core. The surfaces where one shell ends and the next begins are named after the scientists who found them. We know these boundaries only because earthquake waves change speed and direction there.

- Crust-mantle boundary = Mohorovicic discontinuity (Moho).
- Mantle-core boundary = Gutenberg discontinuity.
- By composition the layers are called SIAL (crust), SIMA (mantle), NIFE (core).
- P-waves (primary) pass through solids and liquids both.
- S-waves (secondary) travel only through solids.
- The S-wave shadow zone beyond 103 degrees proves the outer core is liquid.

Interior of the earth and its discontinuities

Upper crust
granitic and light - SIAL

Conrad Discontinuity

Lower crust
basaltic and denser - SIMA
to about 35 km

Mohorovicic (Moho)

Upper mantle
holds the asthenosphere, source of magma
to about 700 km

Repetti Discontinuity

Lower mantle
solid, dense silicate rock
to about 2,900 km

Gutenberg Discontinuity

Outer core
liquid - S-waves stop here
to about 5,150 km

Lehmann Discontinuity

Inner core
solid iron and nickel - NIFE
to 6,371 km

Continental drift – Wegener, 1912

Pangaea, one supercontinent, surrounded by Panthalassa → Splits into Laurasia (north) and Gondwana (south) →

Separated by the Tethys Sea → Fragments drift to their present positions →

Mechanism later explained by Harry Hess – sea-floor spreading

- Circum-Pacific belt = the 'Ring of Fire' – about two-thirds of the world's active volcanoes.
- It is dominated by subduction zones, so it also has the great majority of earthquakes.
- Ojos del Salado (Chile-Argentina) – world's highest volcano.
- Mauna Loa (Hawaii) – largest active volcano.
- Etna – Europe's most active volcano; Stromboli – 'Lighthouse of the Mediterranean'.

YAAD RAKHO

Moho is the mantle's roof, Gutenberg is the core's gate. Roof first, gate deeper – that fixes the order for you.

Mountains – the seven summits and the ranges

This is the single most repeated part of world geography in RAS. The question comes in three shapes: which is the highest peak of continent X, match peaks to continents, and arrange peaks in descending order of height. One table answers all three. Learn the peak, the continent and the range together.

Continent to highest peak

Continent	Highest peak	Height	Range / country
Asia (and the world)	Mount Everest	8,848.86 m	Great Himalaya, Nepal-China border
South America	Aconcagua	about 6,960 m	Andes, Argentina
North America	Denali (formerly Mt McKinley)	6,190 m	Alaska Range, USA
Africa	Mount Kilimanjaro	5,895 m	Tanzania; dormant stratovolcano
Europe	Mount Elbrus	5,642 m	Caucasus, Russia
Antarctica	Vinson Massif	4,892 m	Sentinel Range
Oceania	Puncak Jaya (Carstensz Pyramid)	4,884 m	New Guinea, Indonesia

Descending order of continental highest peaks

Everest → Aconcagua → Denali → Kilimanjaro → Elbrus → Vinson Massif → Puncak Jaya

- K2 / Godwin Austen (8,611 m), Karakoram - world's second highest peak.
- Kangchenjunga (8,586 m) - third highest.
- Mount Kenya (5,199 m) - Africa's second highest.
- Andes (about 7,000 km) - longest mountain range on land.
- Mid-Oceanic Ridge (about 65,000 km) - longest range on earth overall, but under the sea.
- Ural Mountains - conventional boundary between Europe and Asia.
- Atlas - north-west Africa; Drakensberg - South Africa; Great Dividing Range - eastern Australia.

ASKED IN RAS

The direct form. RAS Pre 2018 - which is the highest peak of the African continent? Kilimanjaro. RAS Pre 2016 - Aconcagua lies in which range? The Andes. RAS Pre 2021 - Vinson Massif is the highest peak of which continent? Antarctica. Same table, asked three different years.

ASKED IN RAS

The twisted form. RAS Pre 2021 also asked the peaks arranged in descending order of height, and RAS Pre 2016 asked which peak-continent pair is NOT correctly matched - the wrong pair was 'Mount Kosciuszko - Europe'. So learn the table in both directions and learn the order, not just the pairs.

TRAP

Three confusions RPSC exploits here. (1) Elbrus is Europe's highest; Mont Blanc (4,808 m, Alps) is only the highest in Western Europe. (2) Puncak Jaya is the highest of Oceania; Kosciuszko (2,228 m) is only the highest on mainland Australia - and it is in Australia, never Europe. (3) Everest is the highest peak, but the Andes, not the Himalaya, is the longest range on land.

Plateaus and plains

A plateau is an upland with a flat top standing above the land around it. The Tibetan Plateau is the one that gets named - highest and largest in the world, average elevation above 4,500 m, lying between the Himalaya and the Kunlun, which is why it is called the 'Roof of the World'. For the exam what matters is the plateau's type and its country.

Plateau to type and country

Plateau	Type / feature	Country
Tibetan	Intermontane; 'Roof of the World'	China
Bolivian	Intermontane	Bolivia
Anatolian	Intermontane	Turkey
Deccan	Lava / volcanic	India
Columbia	Lava / volcanic	USA
Colorado	Dissected - the Grand Canyon is cut into it by the Colorado river	USA
Katanga (Shaba)	Copper-rich	DR Congo
Meseta	Continental block	Spain
Mato Grosso	Continental block	Brazil
Loess Plateau	Wind-deposited silt	China
Patagonian	Rain-shadow plateau east of the Andes	Argentina
Chotanagpur	Continental plateau	India

- Plateau types to recall: intermontane, lava, dissected, continental, piedmont, rain-shadow.
- Plains are classified as structural, depositional (alluvial, flood, delta, loess) and erosional (peneplain).
- Named plains: Great Plains (North America), North European Plain, West Siberian Plain.
- Also the Amazon plain (South America), the Pampas plain (Argentina) and the Indo-Gangetic plain (India).
- Depositional plains carry the world's densest populations - they are flat, watered and fertile.

Deserts of the world - the matching RPSC actually asked

RAS Pre 2024 put its world-geography matching question on deserts. That makes this the highest-value table in the chapter. Do not learn deserts as a list; learn each desert with its continent, and then with its country. A desert is any area with very low precipitation - which is why the polar ice sheets count as deserts too.

Deserts of Africa and Asia

Desert	Continent	Country / region
Sahara	Africa	North Africa - the largest hot desert, over 8.5 m sq km
Kalahari	Africa	Southern Africa - Botswana, Namibia, South Africa
Namib	Africa	Coastal Namibia - the oldest desert
Gobi	Asia	Mongolia and northern China
Arabian (with the Rub al-Khali)	Asia	Arabian peninsula
Karakum	Asia	Turkmenistan
Kyzylkum	Asia	Uzbekistan - Kazakhstan
Taklamakan	Asia	Xinjiang, China
Thar	Asia	India - Pakistan
Kavir	Asia	Iran

Deserts of the Americas and Australia

Desert	Continent	Country / feature
Atacama	South America	Chile - the driest desert on earth
Patagonian	South America	Argentina - largest desert of the Americas
Great Basin, Sonoran, Mojave, Chihuahuan	North America	USA and Mexico
Great Victoria	Australia	Largest desert in Australia

Desert	Continent	Country / feature
Great Sandy	Australia	Western Australia

- The two largest deserts on earth are polar: Antarctic (about 13.9 m sq km) and Arctic (about 13.7 m sq km).
- Sahara is the largest HOT desert - that qualifier decides the answer.
- Atacama is kept dry by the cold Peru current offshore and the rain shadow of the Andes.
- Namib is the oldest desert; Patagonian is a rain-shadow desert east of the Andes.

ASKED IN RAS

RAS Pre 2024 asked deserts three ways in the same rotation - the Gobi is spread over Mongolia and northern China; the Patagonian desert lies mainly in Argentina; and a NOT-correctly-matched pair in which 'Namib - South America' was the wrong one. Namib is in southern Africa.

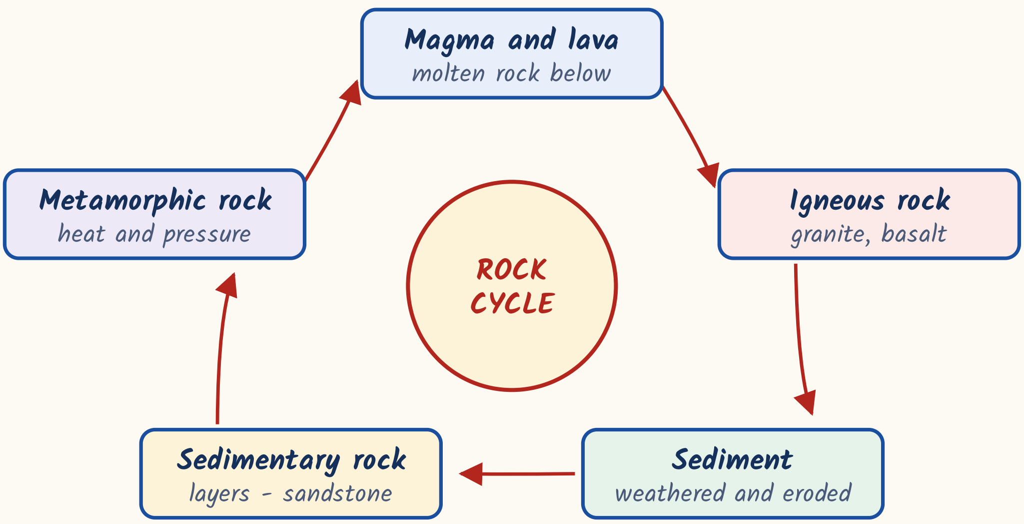
YAAD RAKHO

Four pairs carry the marks - G-P-V-K. Gobi = Asia, Patagonia = South America, Victoria = Australia, Kalahari = southern Africa. And one rule that explains the west-coast deserts: a cold current offshore means a desert on the coast - Peru current with Atacama, Benguela with Namib, Canary with the western Sahara.

Landforms of erosion and deposition

Wind shapes the deserts, ice shapes the high mountains and high latitudes. RPSC asks these as 'which one of these is NOT a glacial landform' - so you need to know which agent made which feature, not the process. Remember one thing above all: yardang is wind, not ice.

The rock cycle



Wind (desert) landforms vs glacial landforms

Wind / desert	Glacier / ice
Erosion: yardang (wind-fluted ridge), zeugen, inselberg, ventifact, mushroom rock, pediment	Erosion: cirque, arete, horn, U-shaped valley, hanging valley, fjord (a drowned glacial valley invaded by the sea)
Deposition: barchan (crescent dune, horns pointing downwind), seif or longitudinal dune, loess (wind-blown silt)	Deposition: moraine, drumlin, esker, outwash plain

CURRENT LINK

Current link: the United Nations declared 2025 the International Year of Glaciers' Preservation, and 21 March is now observed every year as World Day for Glaciers (UNESCO and WMO). Glacier retreat is the standard current-affairs peg for cirques, moraines and fjords - as of September 2026 this remains the live campaign.

Atmosphere, winds, currents and the sea

The atmosphere is layered by how temperature behaves with height. All weather sits in the lowest layer; the ozone that protects us sits in the next one up. Above that, pressure belts and the winds between them set the world's climatic pattern - and the named local winds are a favourite matching set.

- Upward order: troposphere - stratosphere - mesosphere - thermosphere (ionosphere) - exosphere.
- Troposphere holds all weather, water vapour and dust; 18 km thick at the equator, only 8 km at the poles.
- Normal lapse rate in the troposphere - 6.5 degrees C fall per km of ascent.
- Ozone layer lies in the stratosphere, not the troposphere.
- Mesosphere is the coldest layer; meteors burn up here.

Layers of the atmosphere and its pauses

Exosphere — thinnest air; merges into space	above about 400 km
Thermosphere (ionosphere) — radio waves are reflected back to earth	about 80-400 km
Mesopause	
Mesosphere — coldest layer; meteors burn up here	about 50-80 km
Stratopause	
Stratosphere — holds the ozone layer; calm, ideal for jets	up to about 50 km
Tropopause	
Troposphere — all weather; temperature falls 6.5 C per km	18 km eq, 8 km pole

CURRENT LINK

Current link: the ozone layer of the stratosphere is the standing example of an environmental treaty that worked. The WMO-UNEP Scientific Assessment of Ozone Depletion finds the layer on course to return to its 1980 values by about 2040 over most of the world, about 2045 over the Arctic and about 2066 over Antarctica, so long as the Montreal Protocol is kept to; 16 September is World Ozone Day. As of September 2026 those are the recovery dates to quote.

- Four pressure belts: equatorial low (doldrums), sub-tropical high (horse latitudes), sub-polar low, polar high.
- Planetary winds blow from high to low pressure: Trade winds, Westerlies, Polar Easterlies.
- The Westerlies of the Southern Ocean are the 'Roaring Forties'.
- Local winds are named, seasonal and regional - they are what get matched in the exam.

Pressure belts and the planetary winds between them

Polar High — 90 N and 90 S
very cold, dense air sinking

Polar Easterlies

Sub-polar Low — 60-65 N and S
rising air; temperate cyclones form here

Westerlies - the Roaring Forties

Sub-tropical High — 30-35 N and S
horse latitudes; descending dry air

Trade Winds

Equatorial Low — 0 - the Equator
the doldrums; calm, convectional rain

Local winds - name, home and nature

Local wind	Where it blows	Nature
Chinook	Eastern slopes of the Rockies	Warm and dry - the 'snow-eater'
Foehn	Alps	Warm and dry
Mistral	Rhone valley, France	Cold
Bora	Adriatic coast	Cold
Sirocco	Sahara to the Mediterranean	Hot and dust-laden
Harmattan	West Africa	Dry; called 'the Doctor' for the relief it brings
Santa Ana	California	Hot and dry
Zonda	Argentina	Warm and dry, off the Andes

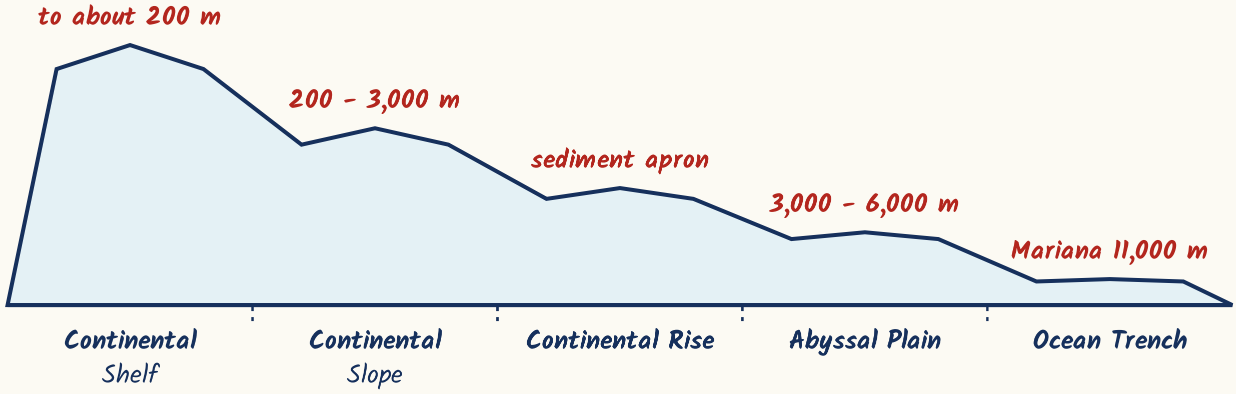
Local wind	Where it blows	Nature
Loo	North India	Hot and dry summer wind

Ocean currents – warm against cold

Warm currents	Cold currents
Gulf Stream	Labrador
North Atlantic Drift	Canary
Kuroshio	Oyashio
Brazil	Peru (Humboldt)
Agulhas	Benguela
East Australian	California
–	West Wind Drift

- Spring tides occur on new moon and full moon, when sun, earth and moon are in line (syzygy).
- Neap tides occur at quadrature, when the sun and moon pull at right angles – the weakest tides.
- Bay of Fundy, Canada – the world's highest tidal range.
- Coral reef types: fringing, barrier and atoll.
- Great Barrier Reef (Coral Sea, off Australia) – the world's largest coral reef system.

Profile of the ocean floor, coast to deep sea



REVISION RECAP

1. Moho separates crust from mantle; Gutenberg separates mantle from core; S-waves stop at the liquid outer core.
2. Wegener 1912 - Pangaea in Panthalassa, splitting into Laurasia and Gondwana across the Tethys; Hess explained the mechanism.
3. Ring of Fire = Circum-Pacific belt = about two-thirds of active volcanoes; Ojos del Salado is the highest volcano.
4. Everest > Aconcagua > Denali > Kilimanjaro > Elbrus > Vinson Massif > Puncak Jaya.
5. Elbrus is Europe's highest; Mont Blanc only Western Europe's; Kosciuszko only mainland Australia's.
6. K2 (8,611 m) second in the world, Kangchenjunga (8,586 m) third; Andes longest range on land, Mid-Oceanic Ridge longest overall.
7. Tibet - highest and largest plateau; Colorado - dissected, Grand Canyon; Katanga - copper, DR Congo; Meseta - Spain.
8. Largest deserts overall are the Antarctic and Arctic polar deserts; Sahara is only the largest hot desert.
9. Gobi-Asia, Patagonia-South America, Great Victoria-Australia, Kalahari-southern Africa: the 2024 matching set.
10. Atacama driest (cold Peru current plus Andean rain shadow); Namib oldest; both lie on west coasts.
11. Yardang, zeugen, barchan and loess belong to wind; cirque, arete, horn, fjord, drumlin and esker belong to ice.
12. Troposphere has all the weather, ozone sits in the stratosphere, mesosphere is the coldest layer.

13. Chinook and Foehn are warm; Mistral and Bora are cold; Harmattan is 'the Doctor'; Sirocco is dusty.
14. Warm - Gulf Stream, Kuroshio, Brazil, Agulhas; cold - Labrador, Canary, Peru, Benguela, Oyashio.
15. Spring tide at syzygy, neap tide at quadrature; Bay of Fundy highest range; Great Barrier Reef largest reef.